# Deepali Banka

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## EDUCATION

**Vellore Institute of Technology** — Chennai, India
*Bachelor of Technology in Electronics and Computer Engineering | GPA: 8.9/10* — Aug 2023 – Jun 2027

## TECHNICAL SKILLS

**Core Research Paradigms**: Signal Processing (SP), Cyber-Physical Systems (CPS), Computer Vision (CV), Real-Time Embedded Systems, Acoustic Deep Learning (Acoustic AI), Defense R&D

**Algorithmic and Machine Intelligence**: Deep Neural Networks (DNNs), Convolutional Neural Networks (CNNs), Sensor Array Fusion, Time-Difference-of-Arrival (TDOA), Statistical Signal Estimation, Multi-Sensor Synchronization

**Languages and Frameworks**: Lingua Franca (LF), Python, C/C++, Java, SQL, MATLAB, PyTorch, OpenCV, Scikit-learn, NumPy, Pandas, Seaborn, Git, Linux/Unix

## RESEARCH AND PROFESSIONAL EXPERIENCE

**Indian Institute of Technology Delhi (IIT Delhi)** — New Delhi, India
*Research Intern, Centre for Applied Research in Electronics (Advisor: Prof. Arun Kumar)* — May 2026 – Till Date

- Conducting core defense research to develop real-time Signal Processing architectures and Deep Neural Network (DNN) algorithms for transient detection and localization (spatial direction-finding) of gunshots.
- Utilizing Time-Difference-of-Arrival (TDOA) matrices to evaluate wave segregation models (muzzle-blast vs. shock-wave).
- Curating acoustic datasets and implementing sensor array fusion to improve DNN resilience against false alarms in noisy environments.

**Nanyang Technological University (NTU)** — Singapore | Remote
*Research Intern, College of Computing and Data Science (Advisor: Prof. Anupam Chattopadhyay)* — Aug 2025 – Till Date

- Conducting advanced systems research to evaluate deterministic execution and temporal semantics in distributed Cyber-Physical Systems (CPS) by implementing Network Time Protocol (NTP)-based time-synchronization frameworks.
- Developing low-latency Reactor models inside Lingua Franca to guarantee microsecond-level precise timing and eliminate race conditions for safe parallelism across multicore and distributed compute architectures. [Project Page]

**Samsung R&D Institute** — Bangalore, India
*Student Researcher* — Jul 2025 – Jun 2026

- Conducted Machine Learning research focused on acoustic de-reverberation methods to remove echo artifacts from corrupted speech and music signals.
- Built data synthesis pipelines to isolate clean time-series audio parameters from highly reverberant ambient noise.

**Cadence Design Systems** — Noida, India
*Summer Intern* — May 2025 – Jun 2025

- Designed and developed a graph-based pattern matching algorithm to automatically recognize recurring patterns in the design.
- Implemented cursor-centric Zoom and Pan features in Allegro System Capture.

**School of Electronics Engineering (SENSE), VIT** — Hybrid
*Summer Intern* — Jun 2024 – Aug 2024

Summer Internship in VLSI Design, offered by the Centre for Nanoelectronics and VLSI Design (CNVD).

- Worked on the design and ASIC implementation of "Fetch Unit" for a pipelined RISC-V processor. [Paper]

**Team MOVIS** (Build custom Drones and Rovers, annual participation in NASA HERC) — Chennai, India
*Technical Team Member* — May 2024 – Jun 2025

- Developed an image stitching program using Python OpenCV to merge images by detecting and matching their feature points. [Project Page]
- Simulated the navigation of a GPS-Denied Autonomous Drone on Martian surface for NASA HERC, 2025.

## PUBLICATIONS AND MANUSCRIPTS

- Deepali Banka and Anupam Chattopadhyay, "From Imperative to Reactive Time-Synchronized Systems: An NTP Case Study", October 2026. [Submitted Manuscript – arXiv:submit/8305651]

## PROJECTS

**Diffusion Purification for Robust Speech Emotion Recognition** [Project Page]
Designing a prosody-guided audio diffusion purifier to defend Speech Emotion Recognition (SER) systems against adversarial attacks.

**Edge-Accelerated Malware Detection via Hybrid FPGA-CPU Deep Learning Pipeline** [Project Page]
Deployed quantized CNNs onto a PYNQ-Z2 FPGA using Quantization-Aware Training (QAT) and fused INT8 kernels to achieve low power, high-speed malware inference at the edge.

**Human Fall Detection and Risk Prediction** [Project Page]
Trained YOLOv8m architecture on a custom-curated vision dataset to predict human fall trajectories and risk parameters.

## ACADEMIC ACHIEVEMENTS AND LEADERSHIP

- Student fellowship, VLSI Design and Test Conference (VDAT), ’24
- Among top 5% participants in Adobe GenSolve, ’24
- Led my team to win the Triad Battle ’25, secured runners-up in AWS DeepRacer ’23, and finished as a finalist in the VITISH ’23 hackathon
- Secured near perfect scores in C, C++ and Python programming tests conducted by Indian Institute of Technology, Bombay, ’24

## COMMUNITY SERVICE AND CO-CURRICULAR

- Certified Senior First-Aid Professional (St. John Ambulance and Indian Red Cross Society) for emergency response management; delivered community programs and voluntary charitable support for underprivileged youth via Aap Ka Prayas NGO and Smile Foundation.
- Long-term training in Indian Classical Dance, won district and state-level dance competitions with multiple competitive choreography awards.